

Trusted Platform Modules: Locksmith in the Basement?

Tech giants are embedding special chips into their systems to ensure your data stays safe, even if you lose your device.

Once upon a time there was a town where people traded very valuable items. But the town had notorious thieves that kept stealing these items. So the people built a special lockbox and kept the items in them for trading. This was a strong box and could not be broken or destroyed to get the contents inside it. The only way to do that was by unlocking it. The box had two keyholes and did not have any fixed keys. A locksmith could make special keys for the box in pairs — if one of the keys in the pair was used to lock the box, only the second key in the same pair could unlock it.

This lockbox was used widely and it worked well. People kept one key of the pair outside their homes in a publicly visible place. Anyone who wanted to give them something made a copy of that key. They then kept that item in the person's lockbox, used this copied key to lock it, and handed over the box. The second key of the pair was kept by the owner securely and not given to anyone, and only he or she could unlock the box with this second key. The thieves stopped stealing these lockboxes because they were unable to open them without the correct keys.

This worked very well for ages. But then some thieves broke into the homes of people and started stealing the second key that was with the owner. And once they got the key, they stole the contents from the lockboxes. This became a menace in no time as the thieves started stealing the keys from more and more people, who started losing the valuables kept in the lockboxes.

To overcome this challenge, people in the town started making vaults to keep their keys safely. These vaults were unbreakable and strong so the thieves could not steal the keys. The second key was kept in the vault and taken out only when needed.

But this, too, did not work for long because the thieves were able to find the keys to the vault to unlock it. Then they stole the keys from the vault!

The people in the town got together and decided to make a very special vault in the basement of their houses — a vault that was unbreakable and strong, and didn't have a door. It was sealed and only had a small window. Everyone in the town had a trusted locksmith to make their keys. When they were making the vaults, they asked their trusted locksmiths to go inside it, and then sealed the vault. The locksmiths lived inside the vault. When a new set of keys was needed, the locksmith made a pair but gave the owner only one key to be kept outside the house. The other key was kept by the locksmith in the vault. And whenever the second key was needed to unlock a box, the lockbox would be passed through the



Figure 1: A special lock box with two keys (AI generated picture)